

- Basic Counting Principle $\begin{cases} \text{Sum} \\ \text{Product} \end{cases}$
- Permutation
- Combination
- Pigeonhole principle
- Recurrence Relation.
- Generating function.

Sum Rule

If a task can be done either in n_1 ways or in n_2 ways where the two sets of ways are disjoint then the total number of ways to complete the task is $n_1 + n_2$.

Ex:- If we roll two dice find the number of ways to get sum 7 or 11.

$$\text{Sum } 7 = (1,6)(2,5)(3,4)(4,3)(5,2)(6,1)$$

$$\text{Sum } 11 = (5,6)(6,5)$$

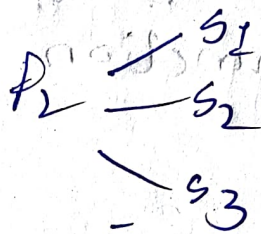
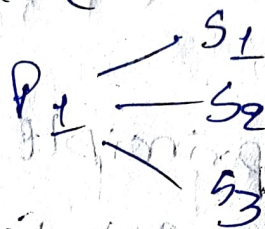
$$6 \oplus 2 = 8$$

Product Rule

Suppose that a procedure can be broken into two tasks if there are n_1 ways to do the first task and n_2 ways to do second task then there are $n_1 \times n_2$ ways to complete the task.

Ex There are 2 pants and 3 shirts how many pairs we can dress-up.

The possible pairs are



Pigeonhole principle

* If n pigeon holes are occupied by $n+1$ or more pigeons then at least one pigeon hole occupied by more than one pigeon hole.

* In general if n objects are placed into k boxes then there is at least one box containing $\lceil \frac{n}{k} \rceil$ objects.

* Suppose a department contains 13 professors then 2 of the professors were born in same month.

A) $\lceil \frac{13}{12} \rceil = \lceil 1.2 \rceil = 2$ professors are born in same month.

* Among 100 people how many are born in same month

$= \lceil \frac{100}{12} \rceil = \lceil 8.3 \rceil = 9$ one pigeon hole must contain 9 members.

Permutation

→ A permutation is an arrangement or ordering of a number of distinct elements. So, there are four types of permutations.

- (i) Permutation without repetition.
- (ii) Permutation with multi-sets
- (iii) Circular permutations.

$$\rightarrow nPr = \frac{n!}{(n-r)!}$$

Permutations without repetition

The total no of permutations of 'n' distinct objects taken all 'n' at a time is n-factorial.

→ No of permutations to arrange the word

$$KLU = 3! = 6 \text{ ways.}$$

Permutation with multi-sets

The total no of permutations of 'n' distinct objects taken 'r' at a time is $\frac{n!}{r!}$ or nPr .

$$= \frac{n!}{(n-r)!}$$

Ex:- How many three letter words can be written by using the word 'THINK'?

$$A) \frac{\text{THINK}}{1 \quad 2 \quad 3} \Rightarrow \frac{5!}{2!}$$

Permutation when repetition of object is allowed.

* The total no of permutations of 'n' objects out of which k objects are same is given by $\frac{n!}{k!}$.

* In general the no of permutations of n objects out of which k_1 are one type, $k_2, \dots, k_n$ are one type is.

$$\rightarrow \frac{n!}{k_1! k_2! \dots k_n!}$$

Ex:- How many words can be found using the letters 'MATHEMATICS'.

no of permutations = $\frac{11!}{2!2!2!}$

② How many different outfits if you have 3 pants and 2 pair of shirts.

A) $P_1 \begin{matrix} s_1 \\ s_2 \end{matrix}$

$P_2 \begin{matrix} s_1 \\ s_2 \end{matrix}$

$P_3 \begin{matrix} s_1 \\ s_2 \end{matrix}$

③ If 5 colours are used to paint 26 doors, show that atleast 6 doors have same colour.

A) $\left\lceil \frac{26}{5} \right\rceil = \left\lceil 5.2 \right\rceil = 6$ pigeons in one hole.

If 7 cars carry 26 passengers prove that at least one car must have 4 or more passengers.

A) $\left\lceil \frac{26}{7} \right\rceil = \left\lceil 3.7 \right\rceil = 4$

B) Show that 50 books in a library contain 27551 pages, one of book must have at least 552 pages

A) $\left\lceil \frac{27551}{50} \right\rceil = \left\lceil 551.02 \right\rceil = 552$

B) How many ways can you arrange BANANAA. If the two A's must be together.

A)
$$\frac{5!}{2!} = \frac{120}{2} = 60$$

Session - 16

B) The chairs of an auditorium are to be labelled with an uppercase english letter followed by a positive integer. What is the largest number of chairs that can be labelled.

A) Total 2600 ways is the largest number of chair.
As 26 alphabets x 100 integers.

② How many ways the letters in the word CAT can be arranged.

A) $CAT = 3! = 6$

③ How many permutations of the letters ABCDEFGH contain the string 'ABC'.

A) $\underbrace{ABC}_{1} DEFGH_{23456}$

$\approx 6! = 720$ ways.

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④ Suppose that we want to buy a computer from one of two A_1 and A_2 and have 12 and 18 of different models respectively. Then how many models are there all together to choose from

A) $A_1 - 12$
 $A_2 - 18$

Total no of models from A_1 or A_2
 $= 12 + 18$
 $= 30$

⑤ Find the number of permutations of the word 'MISSISSIPPI'.

$= \frac{11!}{4! 2! 4!}$

- ⑥ Suppose that there are 8 runners in the race. The winner receives gold, second place receive silver and third place receive bronze medal. How many ways are these medals, if all the possible outcomes of the race can occur and there are on tie's.

A)
$$nPr = \frac{n!}{(n-r)!}$$

$$8P_3 = \frac{8!}{(8-3)!} = \frac{8!}{5!}$$

Combination

→ A combination is a selection of 'r' items from a set that has 'n' distinct elements and is denoted by nCr , where

$$nCr = \frac{n!}{(n-r)!r!} \quad (\text{or } \frac{1}{r!} nPr)$$

Session - 17

- ① A committee of 8 students is to be selected from a class consisting of 19 freshman and 34 seniors. In how many ways can three freshman and five seniors be selected. In how many ways can a committee with exactly one freshman be selected. In how many ways can a committee with at most two freshman be selected.

A) Freshman = 19

Seniors = 34

no of selected students = 8

(i) No of ways = ${}^{19}C_3 \times {}^{34}C_5$

$$= \frac{19!}{16! 3!} \times \frac{34!}{(29!) 5!}$$

(ii) No of ways = ${}^{19}C_1 \times {}^{34}C_7$

$$= \frac{19!}{18! 1!} \times \frac{34!}{27! 7!}$$

(iii) No of ways = of or 1f or 2f.

So, ${}^{19}C_0 \times {}^{34}C_8 + {}^{19}C_1 \times {}^{34}C_7 + {}^{19}C_2 \times {}^{34}C_6$

② A club consists of 6 men and 7 women.
 (i) In how many ways can we select a committee of three men and four women.

(ii) 4 Persons in which atleast one women.

(iii) 4 Persons that has atleast one men.

(iv) 4 Persons which has atmost one men.

A)
 (i) ${}^6C_3 \times {}^7C_4$

(ii) 1w or 2w or 3w or 4w
 $= {}^6C_3 \times {}^7C_1 + {}^6C_2 \times {}^7C_2 + {}^6C_1 \times {}^7C_3 + {}^6C_0 \times {}^7C_4$

(iii) 1m or 2m or 3m or 4m
 $= {}^6C_1 \times {}^7C_3 + {}^6C_2 \times {}^7C_2 + {}^6C_3 \times {}^7C_1 + {}^6C_4 \times {}^7C_0$

(iv) 0M or 1M

$$= 0M \& 2W \text{ or } 1M \& 3W$$
$$= 6C_0 \times 7C_4 + 6C_1 \times 7C_3$$

(3) In how many ways out of class of 30 can 20 students be selected for a extracurricular activity. If Rama refuses to be selected. Raja insist on being selected. Gopal and Govind insist on being selected.

A) (i) Rama refuses to be selected.
 $= {}^{29}C_{20}$

(ii) Raja insist on being selected.
 $= {}^{29}C_{19}$

(iii) Gopal and Govind insist on being selected.
 $= {}^{28}C_{18}$

(4) What is the no of ways of choosing 4 cards from a pack of 52 cards. In how many of these 4 cards from the same suit.

- (i) 4 cards belongs to 4 different suits
- (ii) 2 are red and 2 are black

A) 4H or 4D or 4S or 4C

$$\text{no of ways} = {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4$$

(i) 4 from diff suits = 1H & 1D & 1S & 1C

$$= {}^{13}C_1 + {}^{13}C_1 + {}^{13}C_1 + {}^{13}C_1$$

$$\text{Red} = 26$$

$$\text{Black} = 26$$

$$\text{no of ways} = 2R \& 2B$$

$$= {}^{26}C_2 \times {}^{26}C_2$$

Recurrence Relation

* The process of defining an object in terms of smaller versions of itself is called recurrence.

* In general $n! = n(n-1)!$

$$F(0) = 1 \text{ (base)}$$

$$F(n) = n f(n-1) \text{ (recursion)}$$

Q. Find $f(1), f(2), f(3), f(4)$. If $f(n)$ is defined recursively by $f(0) = 1$ and

$$n = 0, 1, 2, \dots$$

$$\text{(i)} \quad f(n+1) = f(n) + 2$$

$$\text{(ii)} \quad f(n+1) = 3f(n)$$

$$\text{(iii)} \quad f(n+1) = 2^{f(n)} + f(n) + 1$$

$$\text{(iv)} \quad f(n+1) = (f(n))^2 + f(n) + 1$$

$$\text{(i)} \quad f(n+1) = f(n) + 2$$

$$n=0 \Rightarrow f(0+1) = \cancel{f(0)} + 2 = 1 + 2 = 3$$

$$n=1 \Rightarrow f(2) = f(1) + 2 = 3 + 2 = 5$$

$$n=2 \Rightarrow f(3) = f(2) + 2 = 5 + 2 = 7$$

$$n=3 \Rightarrow f(4) = f(3) + 2 = 7 + 2 = 9$$

$$\textcircled{ii} f(n+1) = 3f(n)$$

$$n=0 \Rightarrow f(1) = 3f(0) = 3 \cdot 1 = 3$$

$$n=1 \Rightarrow f(2) = 3f(1) = 3 \cdot 3 = 9$$

$$n=2 \Rightarrow f(3) = 3f(2) = 3 \cdot 9 = 27$$

$$n=3 \Rightarrow f(4) = 3f(3) = 3 \cdot 27 = 81$$

$$\textcircled{iii} f(n+1) = 2^{f(n)}$$

$$n=0 \Rightarrow f(1) = 2^{f(0)} = 2^1 = 2$$

$$n=1 \Rightarrow f(2) = 2^{f(1)} = 2^2 = 4$$

$$n=2 \Rightarrow f(3) = 2^{f(2)} = 2^4 = 16$$

$$n=3 \Rightarrow f(4) = 2^{f(3)} = 2^{16}$$

$$\textcircled{iv} f(n+1) = [f(n)]^2 + f(n) + 1$$

$$n=0 \Rightarrow f(1) = [f(0)]^2 + f(0) + 1$$

$$= 1^2 + 1 + 1$$

$$n=1 \Rightarrow f(2) = [f(1)]^2 + f(1) + 1$$

$$= 3^2 + 3 + 1$$

$$= 13$$

$$n=2 \Rightarrow f(3) = [f(2)]^2 + f(2) + 1$$

$$= 169 + 13 + 1$$

$$= 183$$

$$n=3 \Rightarrow f(4) = [f(3)]^2 + f(3) + 1$$

$$= 183^2 + 183 + 1$$

$$=$$

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→ A recurrence relation for the sequence a_n is an equation that expresses a_n in terms of one or more of the previous terms in the sequence $a_0, a_1, \dots, a_{n-1}$ for all integers n with $n \geq n_0$.

→ A sequence is called a solution of a recurrence relation if it satisfies the recurrence relation.

② Describe the tower of Hanoi problem and model it as recurrence relation.

A) The tower of Hanoi consists of 3 pegs A, B and C and a certain number of disks of different sizes.

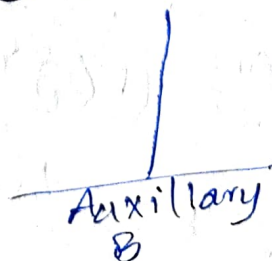
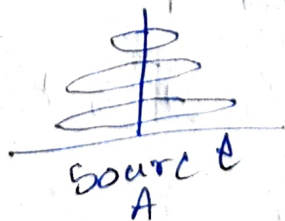
Initially these disks are placed on the first peg 'A' in order of size with the largest at the bottom.

① Only one disk can be moved at a time.

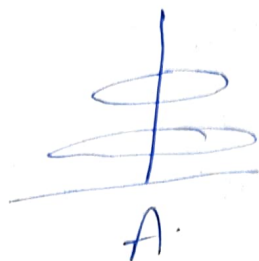
② Largest disk cannot be placed on the top of smaller disk.

③ We can use Auxiliary tower (Peg B) for temporary storage of disk.

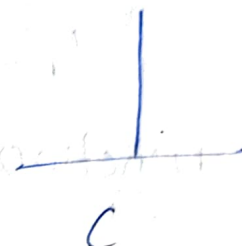
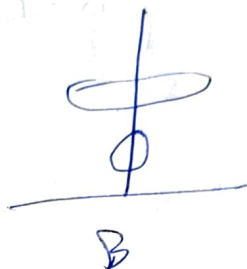
Step 0:-



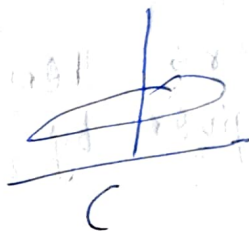
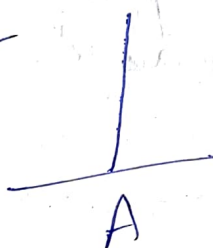
Step 1:-



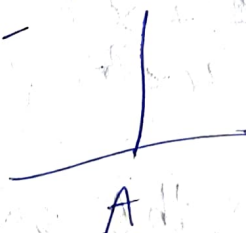
Step 2:-



Step 3:-



Step 4:-



- Transfer the top two disk to the middle peg. Requiring 2 moves.
- Move the largest disk to the 3rd peg requiring one move.
- Transfer the other two disk on the largest disk. Requiring 2 moves.

→ That gives the transferring of 'n' disk.

- Transfer the top $(n-1)$ smaller peg on the middle disk i.e; $n-1$
- Move the largest disk to the 3rd peg i.e; one move.

iii) Again transfers the $n-1$ smallest into the largest. i.e; T_{n-1} moves.

$$* T_n = T_{n-1} + 1 + T_{n-1}$$

$$* T_n = 2 T_{n-1}$$

Generating function

→ If $a_0, a_1, \dots, a_n$ is a sequence of real numbers then its generating function $G(x)$ is given by $G(x) = \sum_{k=0}^{\infty} a_k x^k$.

Note:-

$$G(x) = \begin{cases} (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots \\ (1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 \\ (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 \end{cases}$$

③ Write the generating function in the sequence $\{a_n\}$ with (i) $a_k = 3$ (ii) $a_k = k+1$ (iii) $a_k = 2^k$

A) $a_k = 3$

$$G(x) = \sum_{k=0}^{\infty} a_k x^k$$

$$= \sum_{k=0}^{\infty} 3 \cdot x^k$$

$$= 3 \cdot \sum_{k=0}^{\infty} x^k$$

$$= 3 [1 + x + x^2 + x^3 + \dots]$$

$$= 3(1-x)^{-1} = \frac{3}{1-x}$$

$$a_k = k+1$$

$$G(a_k, x) = \sum_{k=0}^{\infty} a_k \cdot x^k$$

$$= \sum_{k=0}^{\infty} (k+1) \cdot x^k$$

$$= 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$= (1-x)^{-2}$$

$$= \frac{1}{(1-x)^2}$$

$$a_k = 2^k$$

$$G(a_k, x) = \sum_{k=0}^{\infty} a_k x^k$$

$$= \sum_{k=0}^{\infty} 2^k x^k$$

$$= 1 + 2x + 2^2 x^2 + 2^3 x^3 + 2^4 x^4 + \dots$$

$$= [1 + 2x + 2^2 x^2 + 2^3 x^3 + \dots]$$

$$= [1 + (2x) + (2x)^2 + (2x)^3 + \dots]$$

$$= (1-2x)^{-1}$$

$$= \frac{1}{1-2x}$$

4) Determine the generating function for the sequence 1, 1, 1, ...

A) Given sequence is a constant sequence $a_k = 1$

$$G(a_k, x) = \sum_{k=0}^{\infty} a_k x^k$$

$$= \sum_{k=0}^{\infty} 1 \cdot x^k$$

$$= 1 + x + x^2 + x^3 + \dots = (1-x)^{-1} = \frac{1}{1-x}$$

5) Find G.f for the sequence 1, 2, 3, 4, ...

$$a_k = k+1$$

$$G(a_k, x) = \sum_{k=0}^{\infty} a_k x^k$$

$$= \sum_{k=0}^{\infty} (k+1) \cdot x^k$$

$$= 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$= (1-x)^{-2}$$

$$= \frac{1}{(1-x)^2}$$

6) Obtain the generating function for the sequence 0, 1, -2, 3, -4, ...

A) 1, 2, 3, 4, ...

$$(-1)^{k+1} \cdot k$$

$$k=0 \quad (-1) \cdot 0 = 0$$

$$k=1 \quad (-1)^2 \cdot 1 = +1$$

$$k=2 \quad (-1)^3 \cdot 2 = -2$$

$$k=3 \quad (-1)^4 \cdot 3 = +3$$

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Solving Recurrence Relation

$$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_k a_{n-k} + f(n)$$

where C_1, C_2 and C_k are constants.

* If $f(n) = 0$ then it is a homogeneous recurrence relation.

* If $f(n) \neq 0$ then it is a non-homogeneous recurrence relation.

* The general solution of the recurrence relation is $a_n = a_n^{(h)} + a_n^{(p)}$.

* Where $a_n^{(h)}$ is a solution of homogeneous recurrence relation and $a_n^{(p)}$ is particular solution of non-homogeneous recurrence relation.

To find $a_n^{(h)}$

Let the given recurrence relation is of the form $a_n + C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_n a_{n-k} = 0$

$$a_n = a_n^{(h)} + a_n^{(p)}$$

Substitute $a_n = r^n$

$$\Rightarrow r^n + C_1 r^{n-1} + C_2 r^{n-2} + \dots + C_n r^{n-k} = 0$$

$$r^{n-k} [r^k + C_1 r^{k-1} + C_2 r^{k-2} + \dots + C_n] = 0$$

$$r^k + C_1 r^{k-1} + C_2 r^{k-2} + \dots + C_n = 0.$$

Homogeneous linear recurrence relation

Step 1:- Write the auxiliary equation of the given equation.

Step 2:- Find the roots of the auxiliary eqⁿ.

Write the general solution of the given eqⁿ i.e. complementary function ($a_n^{(h)}$) and particular integral solution ($a_n^{(p)}$).

Root

C.F (a_n⁽ⁿ⁾)

1. If r_1, r_2 are real & different.

$$C_1 r_1^n + C_2 r_2^n$$

2. If r_1, r_2 are real and same.

$$(C_1 + C_2 n) r^n$$

3. r_1, r_2, r_3 are real and distinct.

$$C_1 r_1^n + C_2 r_2^n + C_3 r_3^n$$

4. r_1, r_2, r_3 are real & same.

$$(C_1 + C_2 n + C_3 n^2) r^n$$

5. If r_1, r_2 are complex conjugate i.e. $a \pm ib$

$$R^n (A \cos n\theta + B \sin n\theta)$$

where $R = \sqrt{a^2 + b^2}$
 $\theta = \tan^{-1} \left(\frac{b}{a} \right)$

Solve these through H.R.R

Session - 19

i) Solve the fibonacci series

A) $f(n) = f(n-1) + f(n-2)$ where $f(0) = 0$, $f(1) = 1$.

$$f(6) = f(5) + f(4) = 5 + 3 = 8$$

$$f(5) = f(4) + f(3) = 3 + 2 = 5$$

$$f(4) = f(3) + f(2) = 2 + 1 = 3$$

$$f(3) = f(2) + f(1) = 1 + 1 = 2$$

$$f(2) = f(1) + f(0) = 1 + 0 = 1$$

let $f(n) = r^n$

$$r^n = r^{n-1} + r^{n-2}$$

$$r^n - r^{n-1} - r^{n-2} = 0$$

$$r^{n-2} [r^2 - r - 1] = 0 ; r^2 - r - 1 = 0$$

$$r = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot (-1)}}{2 \cdot 1}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

$$= \frac{1 + \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2} \quad (\text{which are real \& different})$$

$$f(n) = C_1 \left(\frac{1 + \sqrt{5}}{2} \right)^n + C_2 \left(\frac{1 - \sqrt{5}}{2} \right)^n \quad \text{--- (2)}$$

with $f(0) = 0$
 $f(1) = 1$

$$\text{put } n=0 \Rightarrow f(0) = C_1 \cdot 1 + C_2 \cdot 1$$

$$\Rightarrow 0 = C_1 + C_2 \quad \rightarrow \textcircled{3}$$

$$\text{put } n=1 \Rightarrow f(1) = C_1 \left(\frac{1 + \sqrt{5}}{2} \right) + C_2 \left(\frac{1 - \sqrt{5}}{2} \right)$$

$$1 = C_1 \left(\frac{1 + \sqrt{5}}{2} \right) + (-C_1) \left(\frac{1 - \sqrt{5}}{2} \right) \quad (\because \text{by } \textcircled{3})$$

$$\Rightarrow 1 = C_1 \left[\frac{1 + \sqrt{5}}{2} - \left(\frac{1 - \sqrt{5}}{2} \right) \right]$$

$$= C_1 \left[\frac{\cancel{1} + \sqrt{5} \cancel{1} + \sqrt{5}}{2} \right]$$

$$= C_1 \left[\frac{2\sqrt{5}}{2} \right]$$

$$1 = C_1 \sqrt{5}$$

$$\therefore \boxed{C_1 = \frac{1}{\sqrt{5}}}$$

$$a_n + 3a_{n-1} + 2a_{n-2} = 0 \quad a_0 = 1; a_1 = 0.$$

$$a_n = r^n$$

$$r^n + 3r^{n-1} + 2r^{n-2} \rightarrow \text{least power.}$$

① Solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$ with $a_0 = 1; a_1 = 0$.

A)
$$a_n = 5a_{n-1} - 6a_{n-2}$$

$$a_n - 5a_{n-1} + 6a_{n-2} = 0 \rightarrow \textcircled{1}$$

$$a_n = r^n$$

$$r^n - 5r^{n-1} + 6r^{n-2} = 0$$

$$r^{n-2} [r^2 - 5r + 6] = 0$$

$$r^2 - 5r + 6 = 0$$

$$r^2 - 2r - 3r + 6 = 0$$

$$r(r-2) - 3(r-2) = 0$$

$$\boxed{r = 2, 3} \text{ real \& diff roots.}$$

$$a_n = C_1 2^n + C_2 3^n \rightarrow \textcircled{2}$$

put $n=0 \Rightarrow a_0 = C_1 2^0 + C_2 3^0$

put $n=1 \Rightarrow a_1 = C_1 2^1 + C_2 3^1 \rightarrow \textcircled{3}$

$$= 2C_1 + 3C_2 \rightarrow \textcircled{4}$$

Solving ③ & ④

$$2c_1 + 2c_2 = 2$$

$$2c_1 + 3c_2 = 0$$

$$\begin{array}{r} 2c_1 + 2c_2 = 2 \\ - \quad 2c_1 + 3c_2 = 0 \\ \hline -c_2 = 2 \end{array}$$

$$c_2 = -2$$

$$c_1 + c_2 = 1$$

$$c_1 - 2 = 1$$

$$c_1 = 3$$

$$a_n = r^n - 3 \cdot 2^n - 2 \cdot 3^n$$

Solve the recurrence relation $a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$ with $a_0 = 3$; $a_1 = 6$; $a_2 = 0$; equation is $a_n - 2a_{n-1} - a_{n-2} + 2a_{n-3} = 0$
 $\rightarrow$ ①

A) $a_n = r^n$ is the solution of ①

$$= r^n - 2r^{n-1} - r^{n-2} + 2r^{n-3} = 0$$

$$r^{n-3} [r^3 - 2r^2 - r + 2] = 0$$

$$r^3 - 2r^2 - r + 2 = 0$$

$$(r=1) \Rightarrow 1 - 2 - 1 + 2 = 0$$

$$r=1 \quad \left| \begin{array}{ccc|c} 1 & -2 & -1 & 2 \\ 0 & 1 & -1 & -2 \\ 1 & -1 & -2 & 0 \end{array} \right|$$

$$(r-1)(r^2 - r - 2) = 0$$

$$r-1=0 \quad r^2 - r - 2 = 0$$

$$r^2 + r - 2r - 2 = 0$$

$$r(r+1) - 2(r+1) = 0$$

$$(r-2)(r+1) = 0$$

$$r = 2, -1, 1$$

$$1 - 2 = -1$$

$$1 - 2 = -1$$

put $n=0$

$$a_n = (1 \cdot 1^n) + (2(-1)^n) + (3 \cdot 2^n) \rightarrow (2)$$

$$\Rightarrow a_0 = c_1 + c_2 + c_3$$

$$3 = c_1 + c_2 + c_3 \rightarrow (4)$$

$n=1$

$$a_1 = c_1 \cdot 1 + c_2(-1) + c_3 \cdot 2$$

$$6 = c_1 - c_2 + 2c_3 \rightarrow (5)$$

$n=2$

$$a_2 = c_1 \cdot 1 + c_2(-1)^2 + c_3 \cdot 2^2$$

$$0 = c_1 + c_2 + 4c_3 \rightarrow (6)$$

Solve (4) & (5)

$$c_1 + c_2 + c_3 = 3$$

$$c_1 - c_2 + 2c_3 = 6$$

$$\underline{2c_1 + 3c_3 = 9} \rightarrow (7)$$

Solve (5) & (6)

$$c_1 - c_2 + 2c_3 = 6$$

$$c_1 + c_2 + 4c_3 = 0$$

$$\underline{2c_1 + 6c_3 = 6} \rightarrow (8)$$

Solve ⑦ & ⑧

$$2c_1 + 4c_3 = 9$$

$$2c_1 + 6c_3 = 6$$

$$-2c_3 = 3$$

$$c_3 = -1$$

$$2c_1 + 6c_3 =$$

$$c_1 + 3c_3 = 0$$

$$c_1 - 3 = 0$$

$$c_1 = 3$$

$$c_1 + c_2 + 4c_3 = 0 \Rightarrow 3 + c_2 - 4 = 0$$

$$c_2 = 1$$

L.S. of ①

$$a_n = r^n = 3 \cdot 1^n + 1 \cdot (-1)^n - 1 \cdot 2^n$$

3) Solve the recurrence relation

$$a_n = 4a_{n-1} - 4a_{n-2} \text{ for } n \geq 2 \text{ with}$$

$$a_0 = 2 \text{ \& } a_1 = 2$$

A)

$$a_n - 4a_{n-1} + 4a_{n-2} = 0 \Rightarrow \textcircled{1}$$

$a_n = r^n \rightarrow$ be the solution of $\textcircled{1}$

$$r^n = 4r^{n-1} + 4r^{n-2} = 0$$

$$r^{n-2} [r^2 - 4r + 4] = 0$$

$$r^2 - 4r + 4 = 0$$

$$(r-2)^2 = 0 \Rightarrow r = 2, 2$$

∴ sol of ① is

$$a_n = (C_1 + C_2) 2^n \rightarrow ②$$

$$\text{put } n=0 \Rightarrow a_0 = C_1 \cdot 1 \Rightarrow C_1 = 2$$

$$n=1 \Rightarrow a_1 = (C_1 + C_2) \cdot 2$$

$$2 = (C_1 + C_2) \cdot 2$$

$$C_1 + C_2 = 1$$

$$2 + C_2 = 1$$

$$C_2 = -1$$

G.S of ① is $a_n = r^n = (2-n) 2^n$

④ Solve the recurrence relation

$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$
with initial conditions $a_0 = 1, a_1 = 3, a_2 = 7$.

$$A) a_n - 6a_{n-1} + 11a_{n-2} - 6a_{n-3} \rightarrow ①$$

$$r^{n-3} [r^3 - 6r^2 + 11r - 6] = 0$$

$$r^3 - 6r^2 + 11r - 6 = 0$$

$$r=1 \Rightarrow 1 - 6 + 11 - 6 = 0$$

$$r=1 \begin{array}{r|rrrr} 1 & 1 & -6 & 11 & -6 \\ & 0 & 1 & -5 & 6 \\ \hline & 1 & -5 & 6 & 0 \end{array}$$

$$(r-1)(r^2 - 5r + 6) = 0$$

$$(r-1)(r-2)(r-3) = 0$$

$$r = 1, 2, 3$$

Sol of

$$\text{at } n=1 \Rightarrow a_1 = C_1 \cdot 1 + C_2 \cdot 2^1 + C_3 \cdot 3^1$$

$$3 = C_1 + 2C_2 + 3C_3 \rightarrow (4)$$

$$n=2 \Rightarrow a_2 = C_1 \cdot 1^2 + C_2 \cdot 2^2 + C_3 \cdot 3^2$$

$$7 = C_1 + 4C_2 + 9C_3 \rightarrow (5)$$

Solving (3), (4), (5)

$$C_1 + C_2 + C_3 = 1 \rightarrow (3)$$

$$C_1 + 2C_2 + 3C_3 = 3 \rightarrow (4)$$

$$-C_2 - 2C_3 = -2 \rightarrow (6)$$

$$C_1 + 2C_2 + 3C_3 = 3 \rightarrow (4)$$

$$C_1 + 4C_2 + 9C_3 = 7 \rightarrow (5)$$

$$-2C_2 - 6C_3 = -4$$

$$C_2 + 3C_3 = 2 \rightarrow (7)$$

Solving (6) & (7)

$$C_2 + 2C_3 = 2$$

$$C_2 + 3C_3 = 2$$

$$-C_3 = 0$$

$$C_3 = 0$$

$$(7) \Rightarrow C_2 + 3C_3 = 2 \Rightarrow C_2 + 0 = 2 \Rightarrow C_2 = 2$$

S.S of ① is

$$a_n = r^n = -1 \cdot 1^n + 2 \cdot 2^n + 0 \cdot 3^n$$

$$r^n = 1 \cdot 1^n + 2 \cdot 2^n$$

⑤ Solve the recurrence relation
 $a_n = 2a_{n-1} - 2a_{n-2}$ for $n \geq -2$ with
 $a_0 = 1; a_1 = 2$

A) $a_n - 2a_{n-1} + 2a_{n-2} \rightarrow$ ①

$a_n = r^n$ be the solution of ①

~~①~~

$$r^n = 2r^{n-1} - 2r^{n-2}$$

$$= r^{n-2} [2r^2 - 2r + 2]$$

$$r^2 - 2r + 2$$

$$r = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1}$$

$$= \frac{2 \pm \sqrt{4-8}}{2}$$

$$= \frac{2 \pm \sqrt{-4}}{2}$$

$$= \frac{2 \pm \sqrt{4}i}{2}$$

$$= \frac{2 \pm 2i}{2}$$

$$= 1 \pm i \text{ (in the form of } a \pm ib)$$

$$a_0 = 1 \quad a_1 = 2$$

$$a_n = 2a_{n-1} - 2a_{n-2}$$

~~Ans~~ →

$$a_n = r^n (A \cos n\theta + B \sin n\theta)$$

$$r = \sqrt{a^2 + b^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$

$$\theta = \tan^{-1}\left(\frac{1}{1}\right)$$

$$\tan \theta = 1$$

$$\boxed{\theta = \frac{\pi}{4}}$$

$$a_n \cdot (\sqrt{2})^n (A \cos \frac{n\pi}{4} + B \sin \frac{n\pi}{4})$$

$$n = 0 \Rightarrow a_0 = 1 (A \cdot 1 + B \cdot 0)$$

$$1 = A$$

$$n = 1 \Rightarrow a_1 = \sqrt{2} \left(A \cdot \frac{1}{\sqrt{2}} + B \cdot \frac{1}{\sqrt{2}} \right)$$

$$2 = A + B$$

$$2 = 1 + B \Rightarrow B = 1$$

G.S of (1) is

$$a_n = r^n = (\sqrt{2})^n \left[\cos \frac{n\pi}{4} + \sin \frac{n\pi}{4} \right]$$

Non-Homogeneous Recurrence Relation

→ The equation is of the form $a_n = C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_k a_{n-k} + f(n) \neq 0$.

where $f(n) \neq 0$ is said to be Non-Homogeneous recurrence relation.

→ The General Solution is $a_n = a_n^{(h)} + a_n^{(p)}$.

To find $a_n^{(p)}$

Let $f(n) = d^n$ or $k \cdot d^n$.

(i) If d is not the root of given equation then assume $a_n^{(p)} = P_0 \cdot d^n$.

(ii) If d is the root of the given eqⁿ then assume $a_n^{(p)} = P_0 \cdot n \cdot d^n$.

(iii) If d is the root of the given eqⁿ twice then $a_n^{(p)} = P_0 \cdot n^2 \cdot d^n$.

a. solve the recurrence relation $a_n = 2a_{n-1} + 2^n$ where $a_0 = 0$.

A) Here $f(n) = 2^n = d^n$

G.S of (1) is

$$a_n = a_n^{(h)} + a_n^{(p)}$$

To find $a_n^{(h)}$:-

$$a_n = 2a_{n-1} \quad (2)$$

let $a_n = r^n$ be the sol of (2)

$$r^n = 2r^{n-1}$$

$$r^n - 2r^{n-1} = 0$$

$$r^{n-1} [r - 2] = 0$$

$$r - 2 = 0$$

$$r = 2$$

$$a_n^{(h)} = C_1 \cdot 2^n$$

To find $a_n^{(p)}$:-

$$f(n) = 2^n = d^n$$

$d=2$ is a root of the eqⁿ.

$$\text{then } a_n^{(p)} = P_0 \cdot n \cdot d^n$$

$$= P_0 \cdot n \cdot 2^n$$

$$a_{n-1} = P_0 (n-1) 2^{n-1}$$

$$P_0 \cdot n \cdot 2^n = 2 \cdot P_0(n-1) 2^{n-1} + \frac{1}{2} n$$

$$P_0 \cdot n \cdot 2^n = 2 \cdot P_0(n-1) 2^{n-1} + \frac{1}{2} n$$

$$P_0 \cdot n \cdot 2^n - P_0(n-1) \cdot 2^n = 2^n$$

$$2^n P_0 [n - (n-1)] = 2^n$$

$$P_0 (n - n + 1) = 1$$

$$P_0 = 1$$

$$\therefore a_n^{(p)} = 1 \cdot n \cdot 2^n$$

$$\text{Sol of (2) is } a_n = C_1 \cdot 2^n + n \cdot 2^n$$

$$a_0 = 0$$

$$a_0 = (C_1 \cdot 2^0 + 0 \cdot 2^0)$$

$$a_0 = C_1 + 0$$

$$a_0 = 1$$

② Solve the recurrence relation
 $a_n = 2a_{n-1} + a_{n-2} = 2$
 for $n \geq 2$

$$\rightarrow a_n^{(h)}$$

$$a_n - 2a_{n-1} + a_{n-2} = 0$$

$$a_n = r^n$$

$$r^n - 2r^{n-1} + 2r^{n-2}$$

$$r^{n-2} [r^2 - 2r + 1]$$

$$r^2 - 2r + 1$$

$$\frac{1}{2} (r-1)^2$$

$$r^2 - 1r - 1r + 1$$

$$r(r-1) - 1(r-1)$$

$$r = 1, 1 \text{ (repeated)}$$

$$\text{Sol of (2) is } a_n^{(h)} = (C_1 + C_2 n) 1^n$$

$$\text{Here } f(n) = 2$$

$$= 2 \cdot 1^n$$

$$d = 1$$

dig a root of give n
 equation and is repeated
 twice

$$a_n^{(p)} = P_0 \cdot n \cdot 2^n \cdot d^n$$

$$= P_0 \cdot n^2 \cdot 2^n \cdot 1^n$$

$$a_{n-1} = P_0 (n-1)^2 \cdot 1^{n-1}$$

$$a_{n-2} = P_0 (n-2)^2 \cdot 1^{n-2}$$

$$\Rightarrow P_0 \cdot n^2 \cdot 1^n - 2 \cdot P_0 (n-1)^2 \cdot 1^{n-1} + P_0 (n-2)^2 \cdot 1^{n-2} = 2$$

$$P_0 [n^2 - 2(n-1)^2 + (n-2)^2] = 2$$

$$P_0 [n^2 - 2(n^2 - 2n + 1) + n^2 - 4n + 4] = 2$$

$$P_0 [n^2 - 2(n^2 - 2n + 1) + n^2 - 4n + 4] = 2$$

$$P_0 (2) = 2$$

$$P_0 = 1$$

$$\therefore a_n^{(p)} = 1 \cdot n^2 \cdot 1^n$$

$$= n^2$$

③ Solve the recurrence relation $a_n = 5a_{n-1} + 6a_{n-2} + 42 \cdot 7^n$

$$a_n + 5a_{n-1} + 6a_{n-2} = 42 \cdot 7^n$$

$$a_n = r^n$$

$$r^n + 5r^{n-1} + 6r^{n-2} = 42 \cdot 7^n$$

$$r^{n+2} [r^2 + 5r + 6] = 0$$

$$r^2 + 5r + 6$$

$$r^2 + 2r + 3r + 6$$

$$r(r+2) + 3(r+2)$$

$$r = -3, -2$$

$$a_n = a_n^{(h)} = C_1(-2)^n + C_2(-3)^n$$

(P)

$$f(n) = 42 \cdot 7^n$$

$$= k \cdot d^n$$

$d = 7$ is not a root of the eqⁿ.

$$\text{Assume } a_n^{(p)} = P_0 \cdot d^n$$

$$a_n^{(p)} = P_0 \cdot 7^n$$

$$a_{n-1} = P_0 \cdot 7^{(n-1)}$$

$$a_{n-2} = P_0 \cdot 7^{(n-2)}$$

$$\textcircled{2} \rightarrow P_0 \cdot 7^n + 5P_0 \cdot 7^{(n-1)} + 6P_0 \cdot 7^{(n-2)} = 42 \cdot 7^n$$

$$P_0 \cdot 7^n + 5P_0 \cdot \frac{7^n}{7} + 6P_0 \cdot \frac{7^n}{7^2} = 42 \cdot 7^n$$

$$P_0 = \left[1 + \frac{5}{7} + \frac{6}{7^2} \right] = 42$$

$$P_0 = \left[\frac{49 + 35 + 6}{7^2} \right] = 42$$

$$P_0 \left[\frac{90}{49} \right] = 42$$

$$P_0 = \frac{42 \times 49}{90}$$

$$P_0 = \frac{343}{15}$$

G.S of (I) $\Rightarrow$

$$a_n = (C_1(-2)^n + C_2(-3)^n) + \frac{343 \cdot 7^n}{15}$$

④

$$a_n = 3a_{n-1} + 10a_{n-2} + 8^n$$

$$a_n - 3a_{n-1} - 10a_{n-2} = 8^n$$

G.S of eqⁿ (I) is $a_n^{(h)} + a_n^{(p)}$

To find $a_n^{(p)}$:-

$$a_n - 3a_{n-1} - 10a_{n-2} = 0 \textcircled{2}$$

Let $a_n = r^n$ be the sol. of (2)

$$r^n - 3r^{n-1} - 10r^{n-2} = 0$$

$$r^{n-2} [r^2 - 3r - 10] = 0$$

$$r^2 - 3r - 10 = 0$$

$$r^2 + 2r - 5r - 10 = 0$$

$$(r+2) - 5(r+2) = 0$$

$$(r+2)(r-5) = 0$$

$$r = -2, 5.$$

sol of (1) :-

$$a_n^{(n)} = c_1(-2)^n + c_2(5)^n //$$

$$8^n = d^n$$

$d=8$ is not root of the given eqⁿ.

$$a_n^{(p)} = P_0 \cdot d^n$$

$$= P_0 \cdot 8^n$$

$$a_{n-1} = P_0 \cdot 8^{(n-1)}$$

$$a_{n-2} = P_0 \cdot 8^{(n-2)}$$

(1) $\Rightarrow$

$$P_0 \cdot 8^n - 3 \cdot P_0 \cdot 8^{(n-1)} - 10 \cdot P_0 \cdot 8^{(n-2)} = 8^n$$

$$P_0 \cdot 8^n - 3 P_0 \frac{8^n}{8} - 10 \cdot P_0 \cdot \frac{8^n}{8^2} = 8^n$$

$$P_0 \cdot 8^n \left[1 - \frac{3}{8} - \frac{10}{8^2} \right] = 8^n$$

$$P_0 \left[\frac{14 - 24 - 10}{64} \right] = 1$$

$$P_0 \left[\frac{30}{64} \right] = 1$$

$$P_0 = \frac{64}{30}$$

$$\therefore a_n = a_n^{(p)} = \frac{64}{30} \cdot 8^n //$$

Q. Solve the recurrence relation $f_n = 3f_{n-1} + 10f_{n-2} + 7 \cdot 5^n$

$$f_n - 3f_{n-1} - 10f_{n-2} = 7 \cdot 5^n \rightarrow (1)$$

A) C.S of (1) is $f_n = f_n^{(h)} + f_n^{(p)}$

To find $f_n^{(h)}$:-

$$f_n - 3f_{n-1} - 10f_{n-2} = 0 \rightarrow (2)$$

Let $f_n = r^n$ be the sol of (2)

$$r^n - 3r^{n-1} - 10r^{n-2} = 0$$

$$r^{n-2} [r^2 - 3r - 10] = 0$$

$$r^2 - 3r - 10 = 0$$

$$r^2 + 2r - 5r - 10 = 0$$

$$r(r+2) - 5(r+2) = 0$$

$$r = 5, -2$$

Sol of (1) is

$$f_n^{(h)} = c_1(-2)^n + c_2(5)^n$$

$$f_n = 7 \cdot 5^n$$

$d=5$ it is a root in the given eqⁿ

assume $f_n^{(p)} = P_0 \cdot n \cdot d^n$

$$= P_0 \cdot n \cdot 5^n$$

$$f_{n-1} = P_0(n-1) \cdot 5^{n-1}$$

$$f_{n-2} = P_0(n-2) \cdot 5^{n-2}$$

$$P_0 \cdot n \cdot 5^n - 3P_0(n-1)5^{n-1} - 10P_0(n-2)5^{n-2} = 7 \cdot 5^n$$

$$P_0 \cdot n \cdot 5^n - 3P_0(n-1)5^n - 10P_0(n-2)5^n = 7 \cdot 5^n$$

$$P_0 \cdot 5^n \left[n - \frac{3(n-1)}{5} - \frac{10(n-2)}{5^2} \right] = 7 \cdot 5^n$$

$$P_0 \left[\frac{25n - 15n + 15 - 10n + 20}{25} \right] = 7$$

$$\Rightarrow P_0 \begin{bmatrix} 35 \\ 25 \end{bmatrix} = 7 \quad P_0 = \frac{7 \times 25}{5} = 5$$

$$\therefore f_n = 5n \cdot 5^n \mid \text{s.s of } \textcircled{I} \text{ is } (1-2)^n + 5n \cdot 5^n$$

Session-21

② Determine the general solution of recurrence relation. Solve the recurrence relation using generating function.

A) Generating Function

sequence $\{a_n\}$

$$\textcircled{1} \frac{1}{1-x} = 1+x+x^2+x^3+\dots$$

$$= \sum_{n=0}^{\infty} x^n$$

1

$$\textcircled{2} \frac{1}{1+x} = 1-x+x^2-x^3+\dots$$

$$= \sum_{n=0}^{\infty} (-1)^n x^n$$

$(-1)^n$

$$\textcircled{3} \frac{1}{(1-x)^2} = 1+2x+3x^2+4x^3+\dots$$

$$= \sum_{n=0}^{\infty} (n+1) x^n$$

$(n+1)$

$$\textcircled{4} \frac{x}{(1-x)^2} = x+2x^2+3x^3+4x^4+\dots$$

$$= \sum_{n=0}^{\infty} n \cdot x^n$$

for n

$$\textcircled{5} \frac{1}{1-ax} = 1+ax+(ax)^2+(ax)^3+\dots$$

$$= \sum_{n=0}^{\infty} a^n \cdot x^n$$

a^n

② Apply generating function technique to solve $a_n = 2a_{n-1} + 3a_{n-2}; n \geq 2$. With initial conditions $a_0 = 3$ & $a_1 = 4$.

$$A) a_n - 2a_{n-1} - 3a_{n-2} = 0 \rightarrow \textcircled{1}$$

multiply x^n on both sides

$$a_n x^n - 2a_{n-1} x^n - 3a_{n-2} x^n = 0$$

$$\sum_{n=2}^{\infty} a_n x^n - \sum_{n=2}^{\infty} 2a_{n-1} x^n - \sum_{n=2}^{\infty} 3a_{n-2} x^n = 0$$

$$[a_2x^2 + a_3x^3 + \dots] - 2[a_1x^2 + a_2x^3 + a_3x^4 + \dots] - 3[a_0x^2 + a_1x^3 + a_2x^4 + \dots] = 0$$

$$\Rightarrow [a_0 + a_1x + a_2x^2 + a_3x^3 + \dots - a_0 - a_1x] - 2[a_0x + a_1x^2 + a_2x^3 + a_3x^4 + \dots] - 3x^2[a_0 + a_1x + a_2x^2 + \dots] = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} a_k x^k - a_0 - a_1x - 2x[a_0 + a_1x + a_2x^2 + \dots] + 2a_0x - 3x^2 \sum_{k=0}^{\infty} a_k x^k = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} a_k x^k - a_0 - a_1x - 2x \sum_{k=0}^{\infty} a_k x^k + 2a_0x - 3x^2 \sum_{k=0}^{\infty} a_k x^k = 0$$

$$a_k x^k = 0$$

$$\Rightarrow G(x) - a_0 - a_1x - 2x G(x) + 2a_0x - 3x^2 G(x) = 0$$

$$G(x)(1 - 2x - 3x^2) = a_0 + a_1x - 2a_0x$$

$$= 3 + x - 2 \cdot 3x$$

$$= 3 - 5x$$

$$G(x) = \frac{3 - 5x}{(1 - 2x - 3x^2)}$$

$$G(x) = \frac{5x - 3}{(3x^2 + 2x - 1)}$$

$$3x^2 + 2x - 1$$

$$3x^2 + 3x - x - 1$$

$$3x(x+1) - 1(x+1)$$

$$(3x-1)(x+1)$$

$$G(x) = \frac{5x-3}{(3x-1)(x+1)}$$

$$\frac{5x-3}{(3x-1)(x+1)} = \frac{A}{3x-1} + \frac{B}{x+1} \quad \text{--- (2)}$$

$$5x - 3 \Rightarrow A(x+1) + B(3x-1)$$

$$x = -1 \Rightarrow 5(-1) - 3 = B(-3-1)$$

$$-8 = B(-4)$$

$$B = 2$$

$$x = \frac{1}{3} \Rightarrow \frac{5}{3} - 3 = A\left(\frac{1}{3} + 1\right)$$

$$-\frac{4}{3} = A\left(\frac{4}{3}\right) \Rightarrow A = -1$$

$$G(x) = \frac{-1}{3x-1} + \frac{2}{x+1}$$

$$\frac{1}{1-3x} + \frac{2}{1+x}$$

$$\Rightarrow a_n = 3^n + 2(-1)^n$$

③ Solve the recurrence relation using G.F
 $a_n = 7a_{n-1}$ with $a_0 = 5$ for $n = 1, 2, 3, \dots$

A) $a_n - 7a_{n-1} = 0 \rightarrow \textcircled{1}$

$$a_n - 7a_{n-1}x^n = 0$$

$$\sum_{n=1}^{\infty} a_n x^n - \sum_{n=1}^{\infty} 7a_{n-1} x^n = 0$$

$$\Rightarrow [a_1 x + a_2 x^2 + a_3 x^3 + \dots] - 7[a_0 x + a_1 x^2 + a_2 x^3 + \dots]$$

$$\Rightarrow [a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots] - a_0 - 7x[a_0 + a_1 x + a_2 x^2 + \dots]$$

$$G(x) - a_0 - 7x G(x) = 0$$

$$(1 - 7x) G(x) = a_0$$

$$G(x) = \frac{5}{1-7x} \Rightarrow 5 \left(\frac{1}{1-7x} \right) \left[\because G(x) = \frac{1}{1-ax} \cdot a_n = a^n \right]$$

$$a_n = 5 \cdot 7^n$$

④ Suppose that a valid code word is an n -digit number in decimal notation containing an even number of zeros. Let us denote the number of valid code words of length n . Then the corresponding recurrence relation is
 $a_n = 8a_{n-1} + 10^{n-1}$ for $n = 1, 2, 3, \dots$ with initial condition $a_0 = 1$ using G.F to find an explicit formula for a_n .

A) $a_n - 8a_{n-1} = 10^{n-1} \rightarrow \textcircled{1}$

$$a_n x^n - 8a_{n-1} x^n = 10^{n-1} \cdot x^n$$

$$\sum_{n=1}^{\infty} a_n x^n - \sum_{n=1}^{\infty} 8a_{n-1} x^n = \sum_{n=1}^{\infty} 10^{n-1} \cdot x^n$$

$$\Rightarrow [a_1x + a_2x^2 + a_3x^3 + \dots] - 8[a_0x + a_1x^2 + a_2x^3 + \dots] = x + 10x^2 + 10^2x^3 + 10^3x^4 + \dots$$

$$\Rightarrow [a_0 + a_1x + a_2x^2 + a_3x^3 + \dots] - 8x[a_0 + a_1x + a_2x^2 + \dots]$$

$$= x[1 + 10x + 10^2x^2 + 10^3x^3 + \dots]$$

$$G(x) - a_0 - 8xG(x) = x[1 + 10x + (10x)^2 + (10x)^3 + \dots]$$

$$G(x)(1 - 8x) - a_0 = x(1 - 10x)^{-1}$$

$$G(x)(1 - 8x) = \frac{x}{1 - 10x} + 1$$

$$G(x)(1 - 8x) = \frac{x}{1 - 10x} + 1$$

$$G(x)(1 - 8x) = \frac{x + 1 - 10x}{1 - 10x}$$

$$= \frac{1 - 9x}{1 - 10x}$$

$$G(x) = \frac{1 - 9x}{(1 - 10x)(1 - 8x)}$$

$$\frac{1 - 9x}{(1 - 10x)(1 - 8x)} = \frac{A}{1 - 10x} + \frac{B}{1 - 8x}$$

$$1 - 9x = A(1 - 8x) + B(1 - 10x)$$

$$x = \frac{1}{8}$$

$$1 - \frac{9}{8} = A(1 - 8 \cdot \frac{1}{8}) + B(1 - 10 \cdot \frac{1}{8})$$

$$x = \frac{1}{10}$$

$$1 - \frac{9}{10} = A(1 - 8 \cdot \frac{1}{10}) + B(1 - 10 \cdot \frac{1}{10})$$

$$-\frac{1}{10} = B(-\frac{2}{10})$$

$$B = \frac{1}{2}$$

$$x = \frac{1}{10}$$

$$1 - \frac{9}{10} = A(1 - \frac{8}{10}) + B(0)$$

$$\frac{1}{10} = A(\frac{2}{10})$$

$$A = \frac{1}{2}$$

$$G(x) = \frac{1}{2(1 - 10x)} + \frac{1}{2(1 - 8x)}$$

$$G(x) = \frac{1}{2} a^n + \frac{1}{2} a^n$$

$$= \frac{1}{2} 10^n + \frac{1}{2} 8^n$$

//

⑤ $a_n = 3a_{n-1} + 2$ $a_0 = 1$; $n \geq 1$.

A) $a_n - 3a_{n-1} = 2$

Multiply x^n on both sides.

$$a_n x^n - 3a_{n-1} x^n = 2x^n$$

$$\sum_{n=1}^{\infty} a_n x^n - 3 \sum_{n=1}^{\infty} a_{n-1} x^n = \sum_{n=1}^{\infty} 2x^n$$

$$[a_1 x + a_2 x^2 + a_3 x^3 + \dots] - 3[a_0 x + a_1 x^2 + a_2 x^3 + \dots] = 2[x + x^2 + x^3 + x^4 + \dots]$$

$$= [a_1 + a_2 x + a_3 x^2 + a_4 x^3 + \dots - a_0] - 3x[a_0 + a_1 x + a_2 x^2 + \dots] = 2x[1 + x + x^2 + x^3 + \dots]$$

$$\Rightarrow \sum_{n=0}^{\infty} a_n x^n - a_0 - 3x \sum_{n=0}^{\infty} a_n x^n = 2x(1-x)^{-1}$$

$$G(x) - 1 - 3x G(x) = \frac{2x}{1-x}$$

$$G(x)(1-3x) = \frac{2x}{1-x} + 1$$

$$= \frac{2x + 1 - x}{1-x} = \frac{1+x}{1-x}$$

$$G(x) = \frac{1+x}{(1-x)(1-3x)}$$

$$\frac{1+x}{(1-x)(1-3x)} = \frac{A}{(1-x)} + \frac{B}{(1-3x)}$$

$$1+x = A(1-3x) + B(1-x)$$

put $x=1 \Rightarrow 2 = A(1-3)$
 $A = -1$

$$x = \frac{1}{3} \Rightarrow 1 + \frac{1}{3} \Rightarrow B(1 - \frac{1}{3})$$

$$\frac{4}{3} = B(\frac{2}{3}) \quad \boxed{B=2}$$

$$G(x) = \frac{-1}{1-x} + \frac{2}{1-3x}$$

$$= -1(1-x)^{-1} + 2(1-3x)^{-1}$$

$$= -1[1+x+x^2+x^3+\dots] + 2[1+3x+(3x)^2+(3x)^3+\dots]$$

$$= -1 \sum_{n=0}^{\infty} x^n + 2 \sum_{n=0}^{\infty} (3x)^n$$

$$\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} [-1 + 2 \cdot 3^n] x^n$$

$$a_n = -1 + 2 \cdot 3^n$$



Induktionsbeweis: Sei $a_n = -1 + 2 \cdot 3^n$ die allgemeine Formel für die Koeffizienten a_n der Potenzreihe $G(x)$. Wir zeigen nun, dass diese Formel für alle $n \geq 0$ gilt. Für $n=0$ gilt $a_0 = -1 + 2 \cdot 3^0 = -1 + 2 = 1$, was mit dem konstanten Term in $G(x)$ übereinstimmt. Angenommen, die Formel gilt für ein beliebiges n . Dann gilt für $n+1$:



Wir erhalten $a_{n+1} = -1 + 2 \cdot 3^{n+1} = -1 + 2 \cdot 3 \cdot 3^n = -1 + 6 \cdot 3^n$. Andererseits gilt $a_{n+1} = -1 + 2 \cdot 3^{n+1} = -1 + 2 \cdot 3 \cdot 3^n = -1 + 6 \cdot 3^n$. Dies zeigt, dass die Formel auch für $n+1$ gilt. Nach dem Induktionsprinzip gilt die Formel $a_n = -1 + 2 \cdot 3^n$ für alle $n \geq 0$.

CO-4 Graph Theory

04/11/21

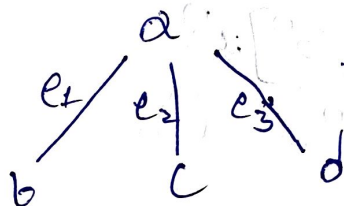
Graph:- A graph 'G' consist of a pair of (V, E) where 'V' is a non-empty finite set whose elements are called vertices and 'E' is another set whose elements are called edges. Such that each edge is associated with pair of vertices.
order of the graph = no of vertices.

* Size of the graph = $|E|$

* Order of the graph = $|V|$

Example:- 'V' = $\{a, b, c, d\}$

$$E = \{(a, b), (a, c), (a, d)\}$$



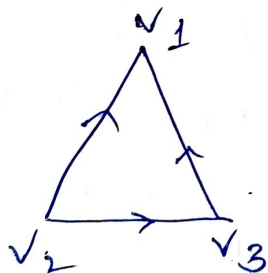
* It is a $(4, 3)$ graph.

Finite graph:- Vertices and edges are countable

Undirected graph:- This is a two way direction graph.



Directed graph:- This is one way direction graph i.e; each edge is given direction.



Self loop:- An edge whose initial & terminal vertex is same.

